

Too few, too many, or just right? Optimizing sample sizes for population-level inferences in animal tracking projects

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S3. Single combination plots

Home range estimation

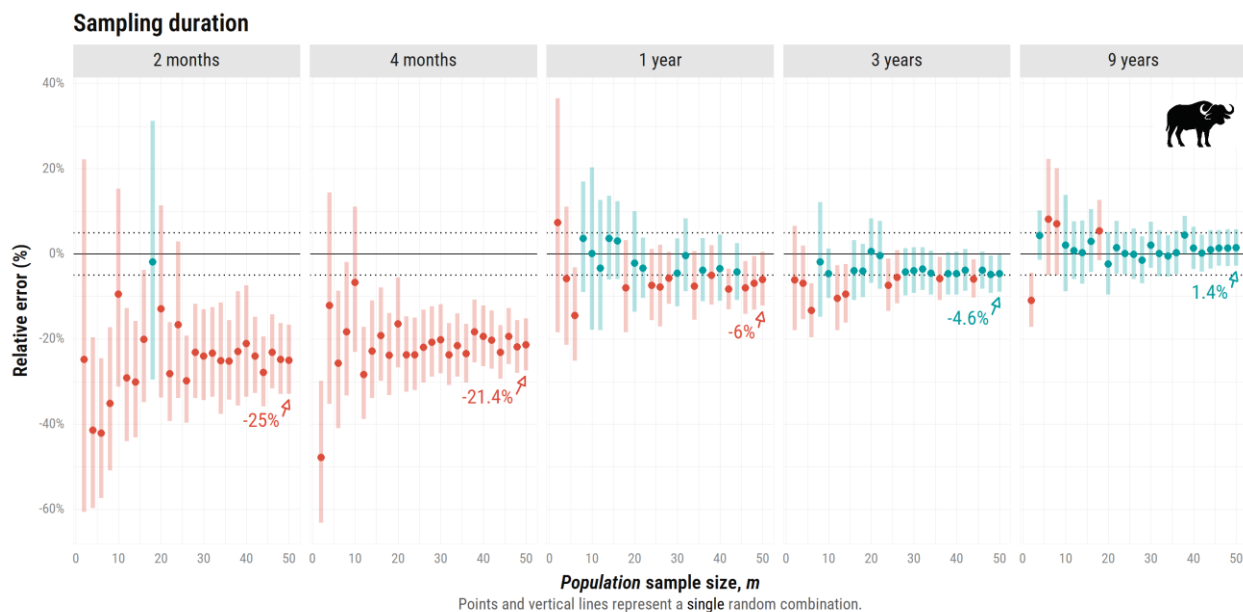


Figure S3.1. Relative error (%) in home range estimates (AKDE) as a function of population sample size (m), ranging from 2 to 50 simulated African buffalos (*Syncerus caffer*). Each vertical facet corresponds to one of five sampling durations: 2 months, 4 months, 1 year, 3 years, and 9 years. The plot shows mean estimates based on a single combination of individuals per m (no resampling); estimates in blue are within the error threshold and in red if estimates fall outside of it. Horizontal dotted lines mark the $\pm 5\%$ error threshold.

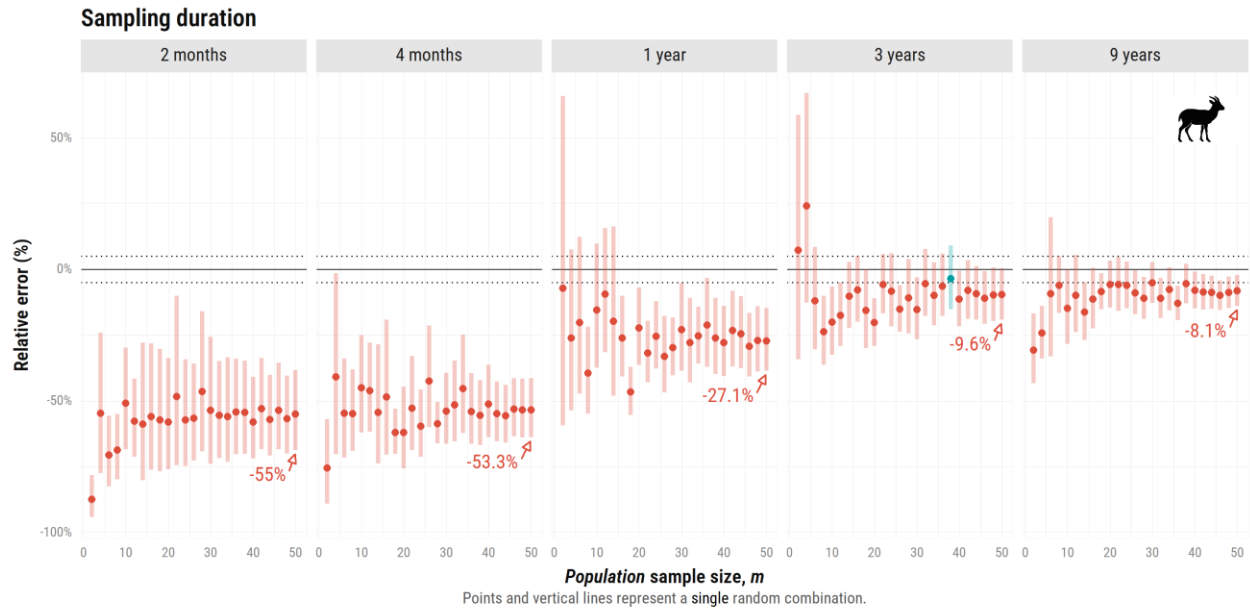


Figure S3.2. Relative error (%) in home range estimates (AKDE) as a function of *population sample size* (m), ranging from 2 to 50 simulated Mongolian gazelles (*Procapra gutturosa*). Each vertical facet corresponds to one of five sampling durations: 2 months, 4 months, 1 year, 3 years, and 9 years. The plot shows mean estimates based on a **single** combination of individuals per m (no resampling); estimates in **blue** are within the error threshold and in **red** if estimates fall outside of it. Horizontal dotted lines mark the $\pm 5\%$ error threshold.

Speed & distance estimation

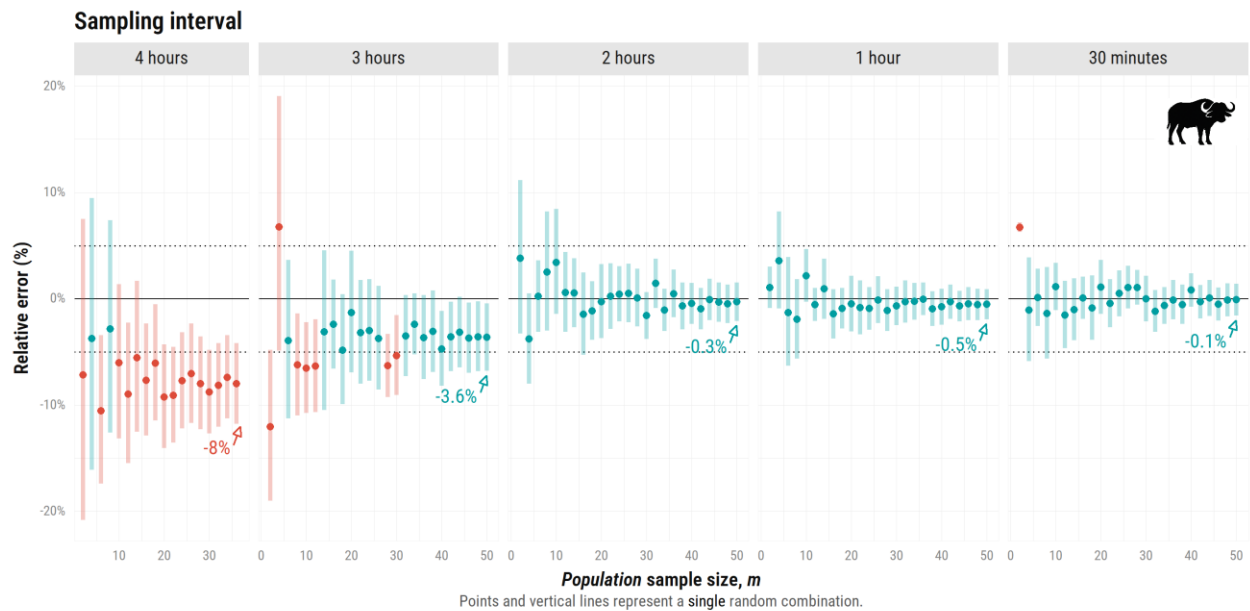


Figure S3.3. Relative error (%) in speed estimates (CTSD) as a function of *population sample size* (m), ranging from 2 to 50 simulated African buffalos (*Syncerus caffer*). Each vertical facet corresponds to one of five sampling intervals: 4 hours, 3 hours, 2 hours, 1 hour, and 30 minutes. The plot shows mean estimates based on a **single** combination of individuals per m (no resampling); estimates in **blue** are within the error threshold and in **red** if estimates fall outside of it. Horizontal dotted lines mark the $\pm 5\%$ error threshold.

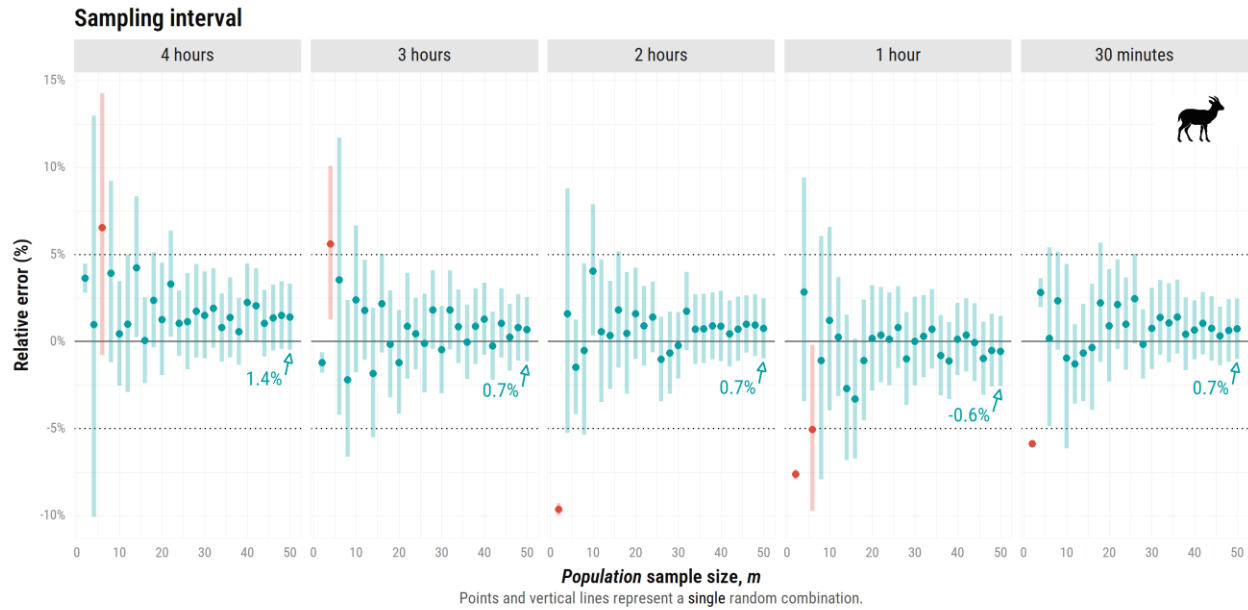


Figure S3.4. Relative error (%) in speed estimates (CTSD) as a function of *population sample size* (m), ranging from 2 to 50 simulated Mongolian Gazelles (*Procapra gutturosa*). Each vertical facet corresponds to one of five sampling intervals: 4 hours, 3 hours, 2 hours, 1 hour, and 30 minutes. The plot shows mean estimates based on a **single** combination of individuals per m (no resampling); estimates in **blue** are within the error threshold and in **red** if estimates fall outside of it. Horizontal dotted lines mark the $\pm 5\%$ error threshold.